

10.2 DC: Potential Dividers

Question Paper

Course	CIEA Level Physics
Section	10. D.C. Circuits
Topic	10.2 DC: Potential Dividers
Difficulty	Medium

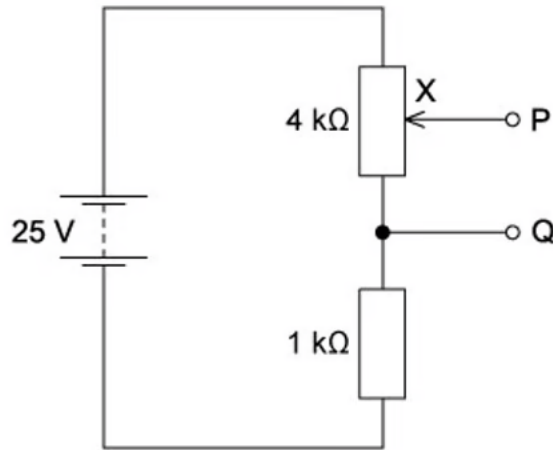
Time allowed: 20

Score: /10

Percentage: /100

Question 1

The diagram shows a potential divider circuit which, by adjustment of the contact X, can be used to provide a variable potential difference between the terminals P and Q.



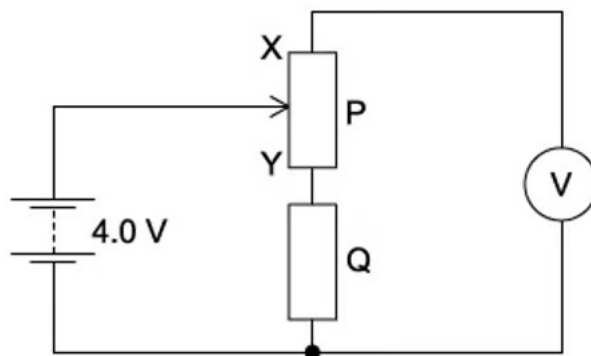
What are the limits of this potential difference?

- A. 0 and 5 V
- B. 0 and 20 V
- C. 0 and 25 V
- D. 5 V and 25 V

[1 mark]

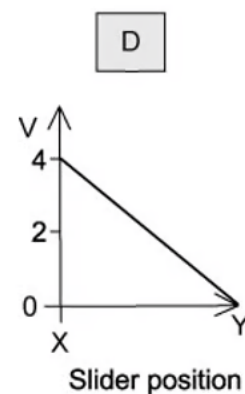
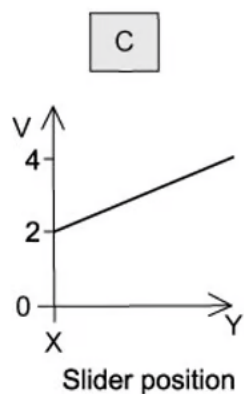
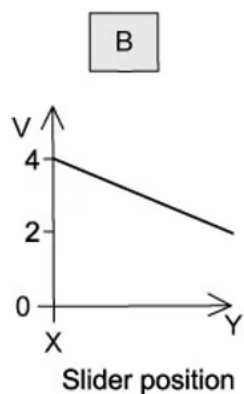
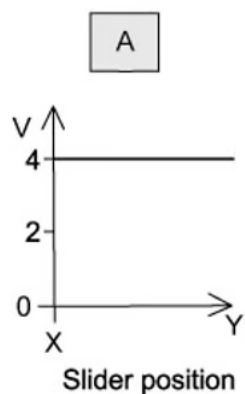
Question 2

In the circuit below, P is a potentiometer of total resistance $10\ \Omega$ and Q is a fixed resistor of resistance $10\ \Omega$. The battery has an e.m.f. of $4.0\ \text{V}$ and negligible internal resistance. The voltmeter has a very high resistance.



The slider on the potentiometer is moved from X to Y, and a graph of voltmeter reading V is plotted against slider position.

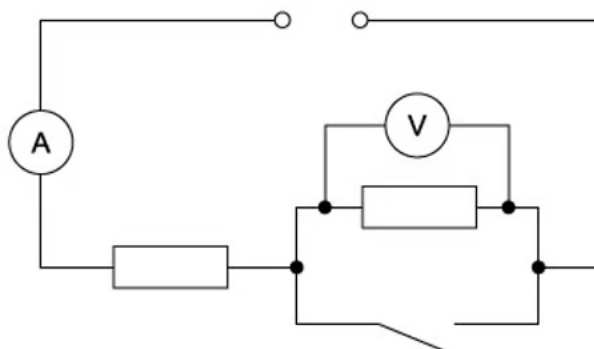
Which graph would be obtained?



[1 mark]

Question 3

In the circuit below, the ammeter reading is I , and the voltmeter reading is V .



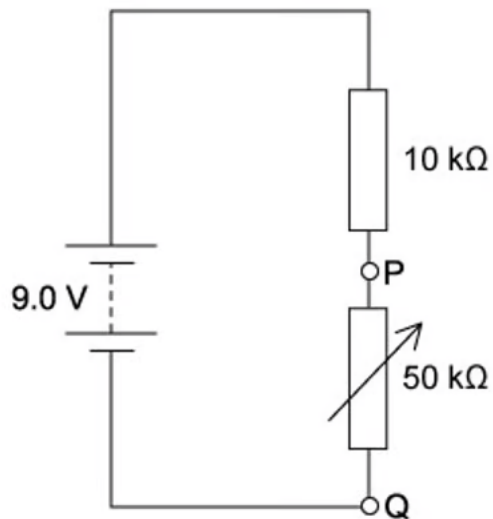
When the switch is closed, which row describes what happens to I and V ?

	I	V
A	decreases	decreases to zero
B	increases	stays the same
C	increases	decreases to zero
D	stays the same	increases

[1 mark]

Question 4

The diagram shows a potential divider connected to a 9.0 V supply of negligible internal resistance.



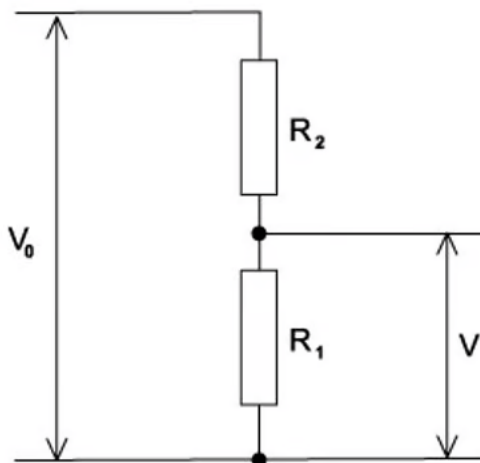
What range of voltages can be obtained between P and Q?

- A. zero to 1.5 V
- B. zero to 7.5 V
- C. 1.5 V to 7.5 V
- D. 1.5 V to 9.0 V

[1 mark]

Question 5

A potential divider consisting of resistors of resistance R_1 and R_2 is connected to an input potential difference of V_0 and gives an output p.d. of V .



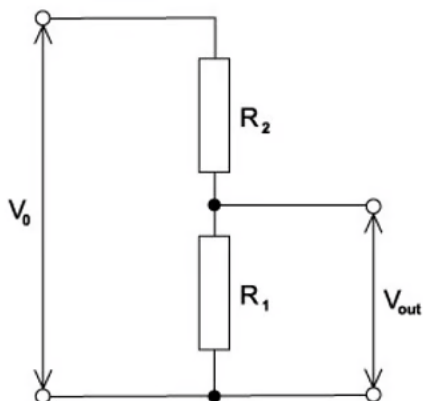
What is the value of V ?

- A. $\frac{V_0 R_1}{R_2}$
- B. $\frac{V_0 R_1}{R_1 + R_2}$
- C. $\frac{V_0 R_2}{R_1 + R_2}$
- D. $\frac{V_0(R_1 + R_2)}{R_1}$

[1 mark]

Question 6

A potential divider consists of resistors of resistance R_1 and R_2 connected in series across a source of potential difference V_0 . The potential difference across R_1 is V_{out} .



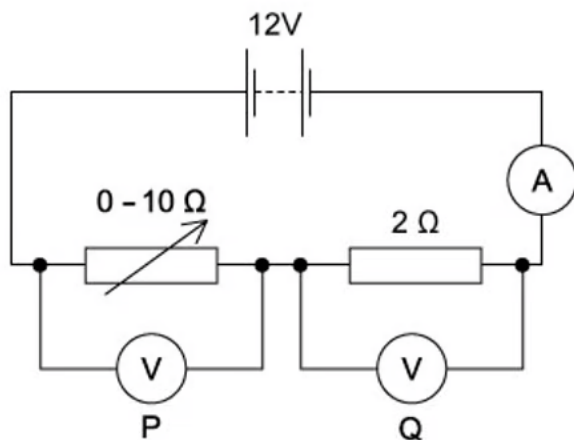
Which changes to R_1 and R_2 will increase the value of V_{out} ?

	R_1	R_2
A	doubled	doubled
B	doubled	halved
C	halved	doubled
D	halved	halved

[1 mark]

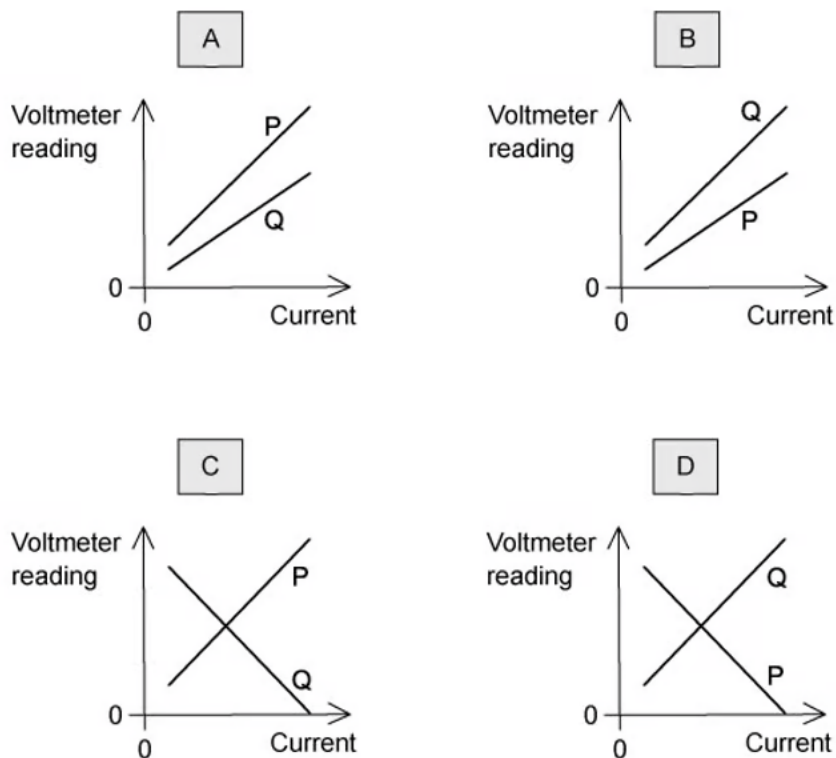
Question 7

A 12 V battery is in series with an ammeter, a 2 W fixed resistor and a $0 - 10\text{ W}$ variable resistor. High-resistance voltmeters P and Q are connected across the variable resistor and the fixed resistor, respectively, as shown.



The resistance of the variable resistor is changed from its maximum value to zero.

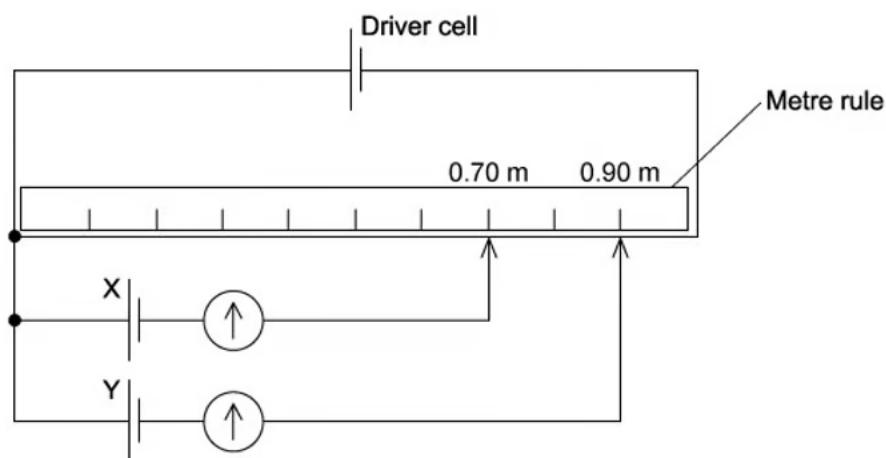
Which graph shows the variation with current of the voltmeter readings?



[1 mark]

Question 8

A potentiometer is used as shown to compare the e. m. f. s of two cells.



The balance points for cells X and Y are 0.70 m and 0.90 m respectively.

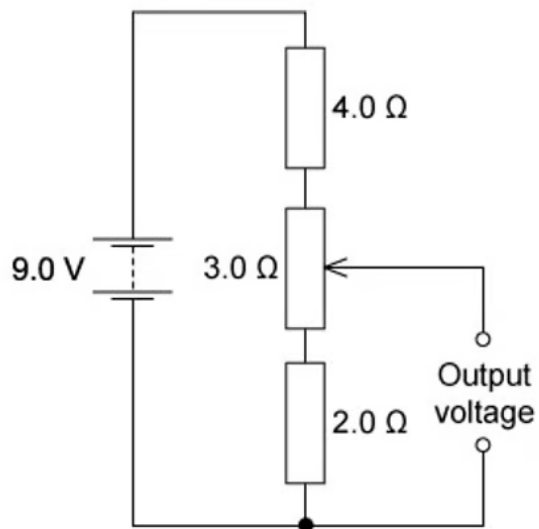
If the e. m. f. of cell X is 1.1 V, what is the e. m. f. of cell Y?

- A. 0.69 V
- B. 0.86 V
- C. 0.99 V
- D. 1.4 V

[1 mark]

Question 9

A potential divider circuit consists of fixed resistors of resistance $2.0\ \Omega$ and $4.0\ \Omega$ connected in series with a $3.0\ \Omega$ resistor fitted with a sliding contact. These are connected across a battery of e.m.f. $9.0\ \text{V}$ and zero internal resistance, as shown.



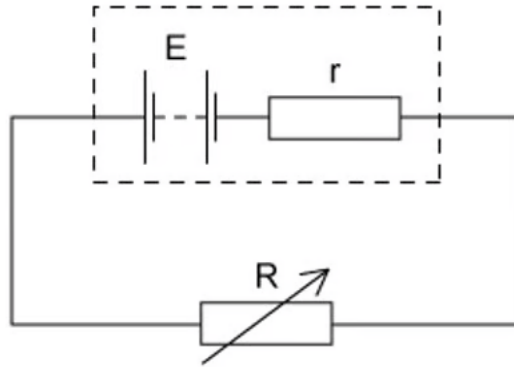
What are the maximum and the minimum output voltages of this potential divider circuit?

	Maximum voltage / V	Minimum voltage / V
A	4.0	2.0
B	5.0	2.0
C	9.0	0
D	9.0	2.0

[1 mark]

Question 10

A battery with e.m.f. E and internal resistance r is connected in series with an external variable resistor.



The value of the external resistance R is slowly increased from zero.

Which statement is correct? (Ignore any temperature effects.)

- A. the potential difference across the external resistance decreases
- B. the potential difference across the internal resistance increases
- C. the power dissipated in r increases and then decreases
- D. the power dissipated in R increases and then decreases

[1 mark]

Model Answers: Medium

1

The correct answer is **B** because:

- The potential divider equation is given by
 - $V_2 = V_{e.m.f} \times \left(\frac{R_1}{R_2 + R_1} \right)$
- Let the resistance of the variable resistor be R_1 and let the resistance of the fixed resistor be R_2
 - R_1 varies from 0–4 k Ω , $R_2 = 1$ k Ω and the e. m. f is $V = 25$ V
- The potential difference between the terminals P and Q is V_2
- The p. d. across a component is proportional to its resistance $V = IR$
 - So, the higher the resistance, the greater the p. d.
- For V_2 to be maximum, the resistance of the variable resistor, R_1 should be 4 k Ω
 - So the maximum p. d. is $V_2 = 25 \times \left(\frac{4}{1+4} \right) = 20$ V
- For V_2 to be minimum, the smallest resistance is chosen, so, R_1 is zero
 - So, the minimum p. d. is $V_2 = 25 \times \left(\frac{0}{1+0} \right) = 0$ V

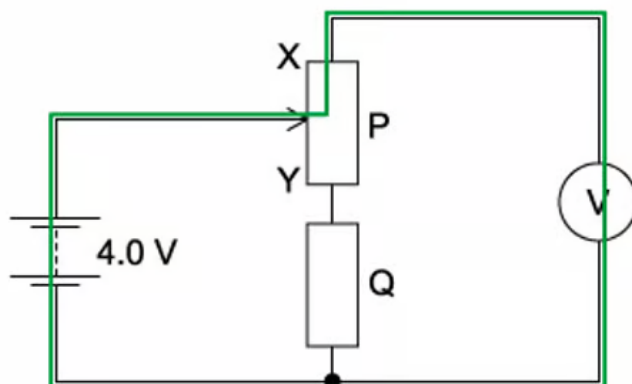
2

The correct answer is **A** because:

- When the slider is at X, where the resistance of the potentiometer is taken to be zero, the voltmeter must read 4 V
- The current flows: from the positive terminal — through the moving jack until the point it is connected on resistor P — from P to end X — through the voltmeter and back to the negative terminal of the supply, as shown in the diagram below
 - Note at no point does the current go through Q
- When the slider is moved to Y, the potentiometer P and the voltmeter are in series
- Still, no current flows through Q — resistor Q can be said to be short-circuited in

both cases, with almost all the resistance in the voltmeter and hence the p.d. across the voltmeter is still 4 V

- Therefore, the voltmeter reading remains 4 V as the slider is moved from X to Y



Exam Tip: In a properly set up circuit, a voltmeter is usually connected in parallel to the component we want to know about. Connecting a voltmeter in series results in the above situation where the voltmeter always reads 4 V

3

The correct answer is **C** because:

- When the switch is opened, current passes through both resistors in the circuit
- Let the resistance of one resistor be R , and we will assume they are of the same resistance
 - When the switch is opened, the total resistance R_T in the circuit $= R + R = 2R$
- Ohm's law states $V = IR$
 - This gives a current $I = \frac{E}{R_T}$ where E is the e.m.f. of the supply
- The total e.m.f in the circuit is divided as p.d. across each of the resistor
- When the switch is closed, the resistor connected in parallel to the wire is short-circuited
- That means, all current would pass through the wire (of negligible resistance) instead of passing through that resistor
 - When the switch is closed, the total resistance R_T in circuit $= R + 0 = R$
- This gives a current I which is now greater than before, since R_T is less
- Since the current I through the resistor in parallel is now zero, the p.d. across it

now also decreases to zero

4

The correct answer is **B** because:

- The potential divider equation is given by
 - $V_2 = V_{e.m.f} \times \left(\frac{R_1}{R_1 + R_2} \right)$
- Let the resistance of the variable resistor be R_1 and let the resistance of the fixed resistor be R_2
 - R_1 varies from 0 – 50 k Ω , $R_2 = 10$ k Ω and the e. m. f is $V = 9.0$ V
- The potential difference between the terminals P and Q is V_2
- The p.d. across a component is proportional to its resistance $V = IR$
 - So, the higher the resistance, the greater the p. d.
- For V_2 to be maximum, the resistance of R_1 should be 50 k Ω
 - So the maximum p. d. is $V_2 = 9 \times \left(\frac{50}{50 + 10} \right) = 7.5$ V
- For V_2 to be minimum, the smallest resistance is chosen, so, R_1 is zero
 - So, the minimum p. d. is $V_2 = 9 \times \left(\frac{0}{10 + 0} \right) = 0$ V
- Therefore, the minimum output of the variable resistor is 0 V, and the maximum output is 7.5 V

5

The correct answer is **B** because:

- We want to find the proportion of output p. d. which is equal to the p. d. across resistor R_1
- For this reason, the R_1 is on the top of the fraction in the potential divider equation
- The potential divider equation is given by
 - $V = V_0 \times \left(\frac{R_1}{R_1 + R_2} \right)$
- Therefore, V is equal to

$$\circ V = \frac{V_0 R_1}{R_1 + R_2}$$

6

The correct answer is **B** because:

- The potential divider equation is given by

$$\circ V_{out} = V_0 \times \left(\frac{R_1}{R_1 + R_2} \right)$$

- In order to increase the value of V_{out} , the value of R_1 should be maximized, while the value of R_2 should be minimized as:

$$\circ V_{out} \propto R_1 \text{ and } V_{out} \propto \frac{1}{(R_1 + R_2)}$$

- Consider option **B**

$$\circ V_{out} \propto \frac{2R_1}{\left(2R_1 + \frac{1}{2}R_2\right)}$$

- Let's use some test values. If $R_1 = 10 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$

$$\circ V_{out} \propto \left(\frac{R_1}{R_1 + R_2} \right) = \left(\frac{10}{10 + 10} \right) = 0.5$$

- So, $2R_1 = 20 \text{ k}\Omega$ and $\frac{1}{2}R_2 = 5 \text{ k}\Omega$

$$\circ V_{out} \propto \frac{2R_1}{\left(2R_1 + \frac{1}{2}R_2\right)} = \left(\frac{20}{20 + 5} \right) = 0.8$$

- Therefore, doubling R_1 and halving R_2 leads to an increase in V_{out}

7

The correct answer is **D** because:

- Since the variable resistor and the fixed resistor are in series, the total e.m.f in the circuit is divided between them

6

The correct answer is **B** because:

- The potential divider equation is given by

$$\circ V_{out} = V_0 \times \left(\frac{R_1}{R_1 + R_2} \right)$$

- In order to increase the value of V_{out} , the value of R_1 should be maximized, while the value of R_2 should be minimized as:

$$\circ V_{out} \propto R_1 \text{ and } V_{out} \propto \frac{1}{(R_1 + R_2)}$$

- Consider option **B**

$$\circ V_{out} \propto \frac{2R_1}{\left(2R_1 + \frac{1}{2}R_2\right)}$$

- Let's use some test values. If $R_1 = 10 \text{ k}\Omega$ and $R_2 = 10 \text{ k}\Omega$

$$\circ V_{out} \propto \left(\frac{R_1}{R_1 + R_2} \right) = \left(\frac{10}{10 + 10} \right) = 0.5$$

- So, $2R_1 = 20 \text{ k}\Omega$ and $\frac{1}{2}R_2 = 5 \text{ k}\Omega$

$$\circ V_{out} \propto \frac{2R_1}{\left(2R_1 + \frac{1}{2}R_2\right)} = \left(\frac{20}{20 + 5} \right) = 0.8$$

- Therefore, doubling R_1 and halving R_2 leads to an increase in V_{out}

7

The correct answer is **D** because:

- Since the variable resistor and the fixed resistor are in series, the total e.m.f in the circuit is divided between them

- The potential difference across each of the resistors may be obtained from the

potential divider equation: $V_P = V \times \left(\frac{R_P}{R_P + R_Q} \right)$

- Let's say R_P is the resistance of the variable resistor and R_Q is the resistance of the fixed resistor

- As R_P is decreased, the p. d. across it decreases ($V = IR$)

- When $R_P = 0$, $V_P = 0$, therefore, the voltmeter reading of P should give zero - option D shows this
- Decreasing R_P should also cause the p.d. across the fixed resistor to increase - this is also shown by option **D**

A, B & C are incorrect because as R_P decreases, the p.d. across it also decreases, whereas these graphs all show the p.d. increasing

8

The correct answer is **D** because:

- At the balance length, the p.d. across the wire is equal to the e.m.f. of the cell
- The p.d. is directly proportional to the length of the wire i.e. $V \propto L$
 - This is because p.d. is proportional to work done which is in turn proportional to distance i.e. $V \propto W \propto d$
- Cell X: $1.1 \text{ V} \propto 0.70 \text{ m}$ **(1)**
- Cell Y: $\text{e.m.f} \propto 0.90 \text{ m}$ **(2)**
- To find the ratio, divide equation **(2)** by **(1)**:
 - $\frac{\text{e.m.f}}{1.1} = \frac{0.90}{0.70}$
 - $\text{e.m.f} = \frac{0.90}{0.70} \times 1.1 = 1.41 \text{ V}$
- Therefore, the e.m.f. of cell Y is 1.4 V

9

The correct answer is **B** because:

- The potential at the top of the circuit (+ terminal) is 9 V, and the potential at the bottom of the circuit (- terminal) is 0 V
- There will be a p.d. across each of the fixed resistors
- Potential divider equation for three resistors: $V_{out} = V \times \left(\frac{R_1 + R_2}{R_1 + R_2 + R_3} \right)$
- The output voltage is maximum at the maximum value of the variable resistor = 3.0Ω
 - Maximum output voltage $= 9.0 \times \left(\frac{2+3}{2+3+4} \right) = 5.0 \text{ V}$
- The output voltage is minimum when the minimum value of the variable resistor is

taken = $0\ \Omega$

$$\circ \text{ Minimum output voltage} = 9.0 \times \left(\frac{2+0}{2+3+4} \right) = 2.0\ \text{V}$$

10

The correct answer is **D** because:

- Ohm's law states $V = IR$, so, as the resistance of a component increases the p. d. across it increases
- So, as the value of external resistance R is increased, the p. d. across it increases, while the p. d. across the internal resistance decreases
- The external resistance R and the internal resistance r form a potential divider
- Power dissipated in external resistor = $I^2 R$
 - Initially, the value of R is zero, so the power dissipated across it is also zero
 - Similarly, if the value of R is infinite, the power dissipated is also zero as the current will then also be zero
- So, the power dissipated must increase and then decrease as R increases from zero
 - As the value of R increases, the current I decreases and hence the p. d. across the external resistor increases
 - The p. d. across the internal resistor r decreases and the power dissipated in r (given by $I^2 R$) decreases
 - The maximum power dissipated across R occurs when $R = r$
- Overall, the only correct statement is that the power dissipated in R increases and then decreases

A & B are incorrect because the sum of p. d. in a circuit is equal to the e.m.f. in the circuit, so as the p. d. of one component rises, the other will fall

C is incorrect because as the value of R increases, the p. d. across the external resistor increases, so the p. d. across the internal resistor decreases and the power dissipation across it also decreases